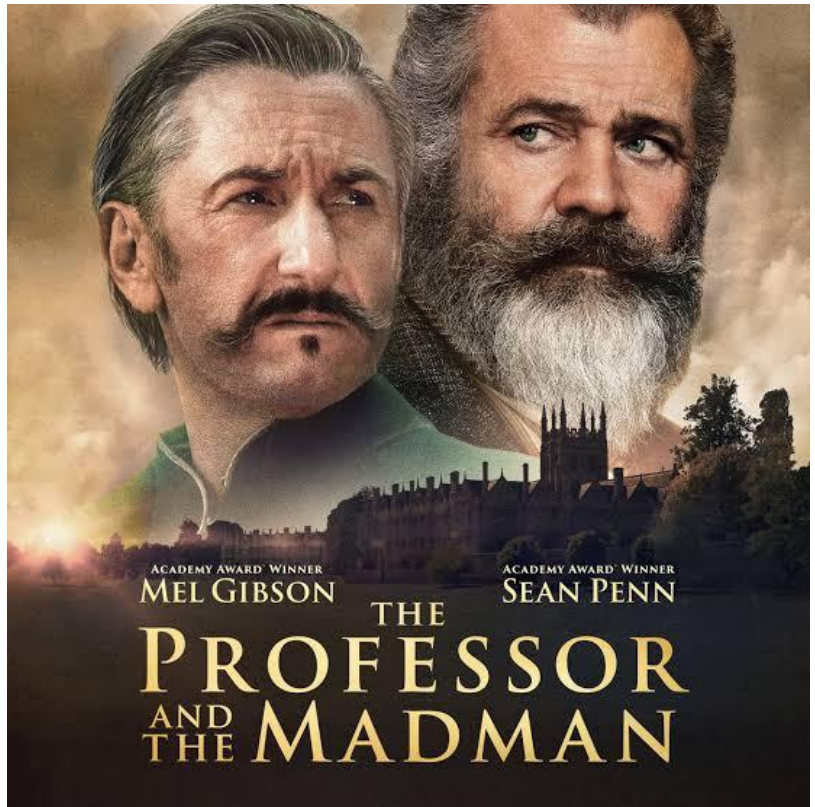
Words

References:

1. Bender: Linguistic Fundamentals for NLP I
2. Bender and Lascarides: Linguistic Fundamentals for NLP II
3. Asher, Kumari: Malayalam

Dictionary

- Oxford English Dictionary
- WordNet
- Wikipedia



Word

- Type and token (class and instance)
- Morpheme
- Meaning
  - Context
  - Intent

Morpheme

- Morphemes are the smallest meaningful units of language.

  Morpheme+s are the small+est mean+ing+ful unit+s of language.

Derivational affixes

Affix	**Example**
-able	*fixable*
-er	*teacher*
-less	*brainless*
anti-	*antidepressant*
de-	*deactivate*

Inflectional affixes

Affix	**Effect/Example**
-s	plural / *cats*
-’s	possessive / *cat’s food*
-s	present tense; 3rd person subject / *jumps*
-ed	past tense / *jumped*
-ed/-en	perfective aspect / *eaten*
-ing	progressive aspect / *jumping*
-er / -est	comparison / *faster; fastest*

Stemming and Lemmatization

- reduce forms of a word to a common base form
  - *am, was, are: to be*
  - *doing, done: do*
  - *park, parks, park's: park*
  - *(parking?)*

https://nlp.stanford.edu/IR-book/html/htmledition/stemming-and-lemmatization-1.html

Word and its forms

varunnavan	‘the male person who is coming’
varunnavaḷ	‘the female person who is coming’
varunnavar	‘the people who are coming’
varunnatə	‘the thing which is coming’
varunnava	‘the things which are coming’
vannavan	‘the male person who came’
vannavaḷ	‘the female person who came’
vannavar	‘the people who came’
vannatə	‘the thing which came’
vannava	‘the things which came’

Representing a word

- Logic
  - a word as a predicate

  person(x)

  food(y)

  eats (p, q, r) : p=person; q=food; r=time

  eats(rahul, chapati, sunday)
- Limitations

Sense

- Senses: possible uses of a given word form
  - Homonymy: same form but different meaning *(bank;?)*
  - Homophones: same pronunciation but different forms (*write-right;?*)
  - Polysemy: related meanings of a word *(school; Beijing;?)*
  - Synonymy: words with almost similar meaning in some contexts (?)
  - Antonyms (?)
  - Hyponymy and Hypernymy
    - *biryani* is a hyponym of *food*
    - *food* is a hypernym of *biryani*
  - Meronymy and Holonymy: part and whole relation
    - *battery* is a meronym of *laptop*, and *laptop* is a holonym of *battery*

https://web.stanford.edu/~jurafsky/slp3/slides/Chapter18_introandsimilarity.pdf

WordNet

- A hierarchically organized lexical database
- IndoWordNet
  - https://www.cfilt.iitb.ac.in/indowordnet/wordnet?langno=0&query=%E0%A4%A8%E0%A4%A6%E0%A5%80

https://wordnet.princeton.edu/
https://www.cfilt.iitb.ac.in/indowordnet/

Semantic role

- Semantic role
  - *eat*
  - *love; hate;*
  - *marriage*
- Structure

$$\begin{bmatrix} \textbf{book} & \\ \text{argstr} : & \begin{bmatrix} \text{arg1} : \textbf{x:ind} \\ \text{arg2} : \textbf{y:abstr} \end{bmatrix} \\ \text{qualia} : & \begin{bmatrix} \text{agentive} : \textbf{write(e,w,y)} \\ \text{telic} : \textbf{read(e,z,y)} \end{bmatrix} \end{bmatrix}$$

Wikipedia

- Title
- InfoBox
- Description
- Hyperlinks
- Categories

Arrival

Theatrical release poster

Directed by	Denis Villeneuve
Screenplay by	Eric Heisserer
Based on	"Story of Your Life" by Ted Chiang
Produced by	Shawn Levy Dan Levine Aaron Ryder David Linde
Starring	Amy Adams Jeremy Renner Forest Whitaker Michael Stuhlbarg Tzi Ma
Cinematography	Bradford Young
Edited by	Joe Walker
Music by	Jóhann Jóhannsson

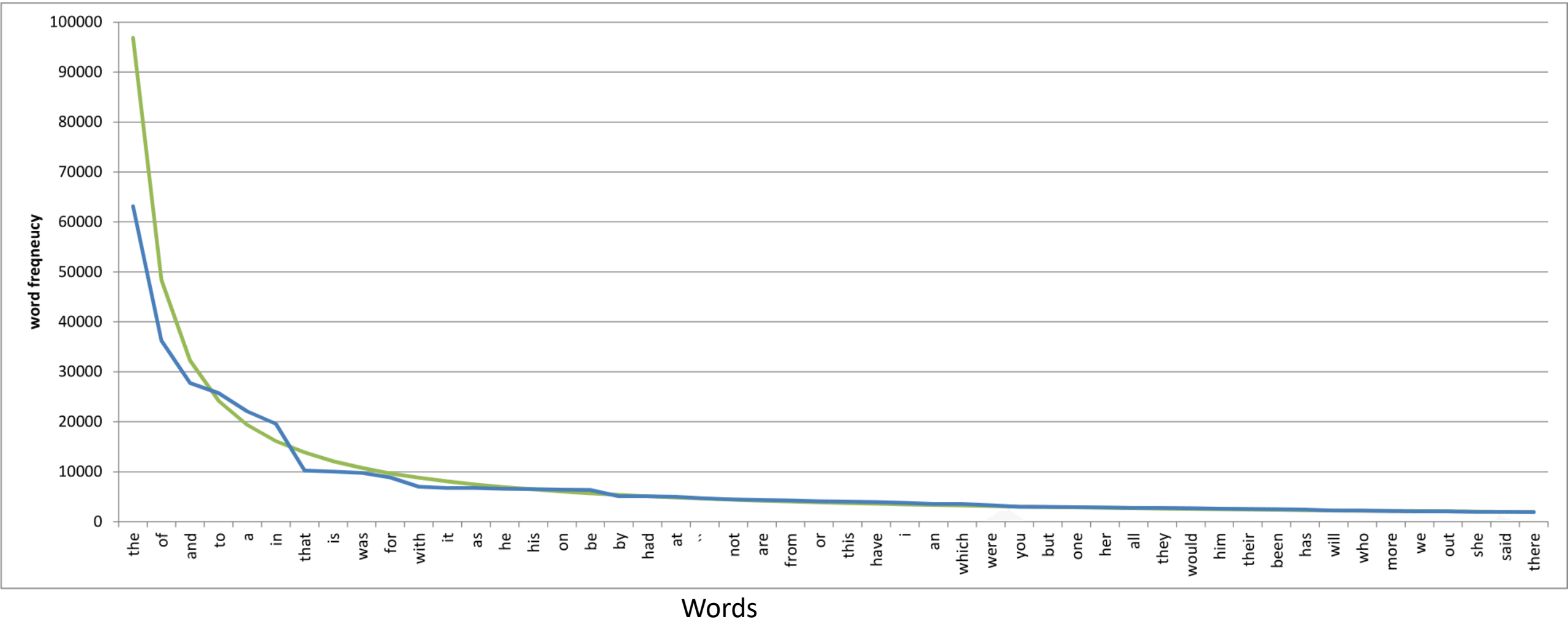
Challenges

- Acquisition
- Representation
- Retrieval

- Meaning and frequency of a word

Zipf's Law

$$\text{word frequency} \propto \frac{1}{(\text{word rank} + \beta)^{\alpha}}$$



Word counts (blue) in the Brown Corpus and the expected word counts according to Zipf's Law (green).

https://shadycharacters.co.uk/2015/10/zipfs-law/

Linguistic aspects

- Most frequent words are more ambiguous as compared to least frequent words
- The more frequent a word is, the shorter it tends to be
  - Abbreviations
- Observed across almost all languages

https://era.ed.ac.uk/handle/1842/33051

Computational aspects

- Word frequencies in a language can be approximated as a distribution
- Importance of a word
  - Domain specific importance
- Disambiguation

Properties of Words

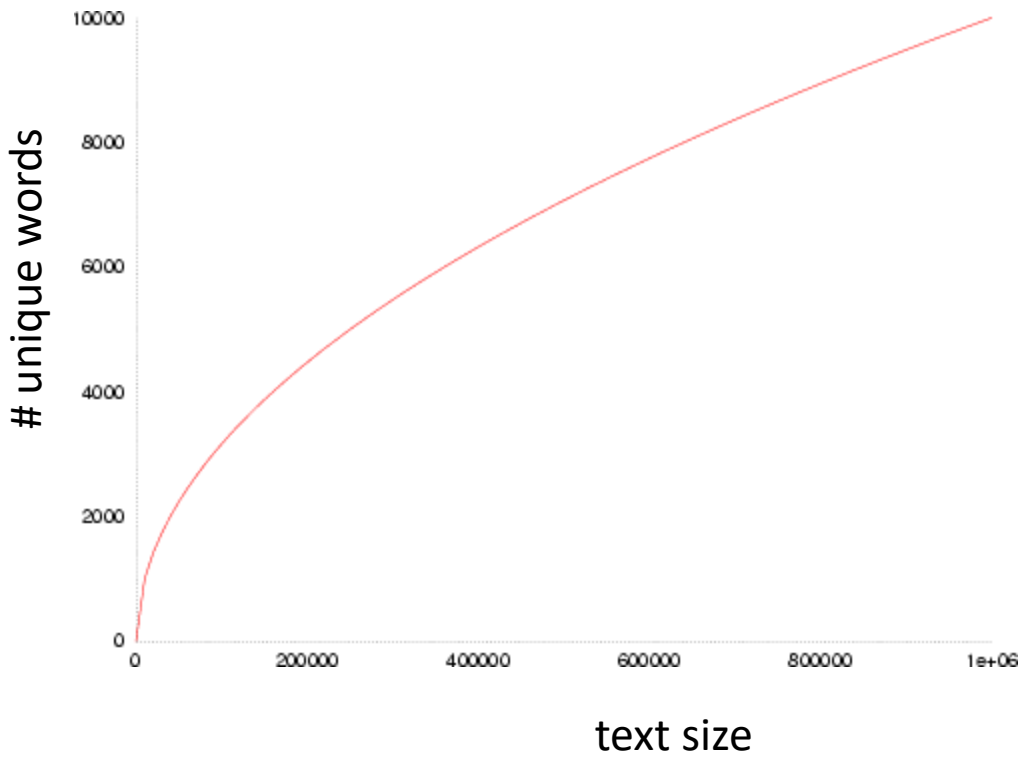
- Word burstiness
  - a word is more likely to occur again in a document if it has already appeared in the document.
  - words that are more informative are also more bursty.
- Do you know a similar phenomenon in the context of computing?
- Heaps' law

Church, K., & Gale, W. A. (1995). "Poisson mixtures". Natural Language Engineering, 1, 163–190.
Doyle, Gabriel, and Charles Elkan. "Accounting for burstiness in topic models." Proceedings of the 26th Annual International Conference on Machine Learning. 2009.

Heaps' Law

- Size of vocabulary: $V(n) = kn^{\beta}$

In English, K is between 10 and 100, β is between 0.4 and 0.6.



https://en.wikipedia.org/wiki/Heaps%27_law

- why the complex language production processes should conform to mathematically concise equations?

Steven T Piantadosi. 2014. Zipf's word frequency law in natural language: A critical review and future directions. Psychonomic bulletin & review, 21(5):1112–1130.